Hold Out Test Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Hold Out Test Data

> Prepare Hold Out Test Data: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: **Freeze** the full evaluation procedure before testing

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a line from a declared **training** candidate set

Reference example, not the only acceptable wording: Teacher explanation: A 5 B 3 C 0; choose m 2 b 1.

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

Look for: Fit mean and median constant baselines on **training** targets

Reference example, not the only acceptable wording: Teacher explanation: Mean 6; median 5.

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Look for: Choose the denominator that answers the question

Reference example, not the only acceptable wording: Teacher explanation: Predicted positives TP+FP=7; reference positives TP+FN=4.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

After seeing test errors, a team adjusts its threshold and reports the improved score on that same test. Rename the set's role and describe an honest final report if no new data exists.

Concept to explain

The final test set must remain untouched until model and threshold choices are finished, or it becomes part of **training** by decision.

Term	Meaning
holdout	reserved unseen evaluation data
contamination	test information influences choices
model selection	choose among candidate models
final estimate	performance measured once after choices freeze

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on final testing and honest **claim**s. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded **claim**s, including justified uncertainty.

Readiness answer

It influenced choices during development.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain hold out test data using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Hold Out Test Data using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Identify contaminated evaluations and repair the workflow.

MODEL IDEA The final test set must remain untouched until model and threshold choices are finished, or it becomes part of **training** by decision.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Team checks test accuracy after every parameter change. Those repeated choices adapt to the test set, contaminating it.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Team checks test accuracy after every parameter change.

Step 2

STEP 2 Those repeated choices adapt to the test set, contaminating it.

Step 3

STEP 3 Use **training** for **fitting** and validation for model or threshold choices.

Step 4

STEP 4 After freezing choices, open the **holdout** once and report the **final estimate** honestly.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

A final-test folder has twenty leaves; its labels are opened weekly to choose models.

Is it still untouched?

Exact explanation to model

Why the result follows

No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

Explain the added worked example

Explain the added worked example · 1

A final test is valuable because it did not guide model choices. **Freeze** the model, **training** pre**process**ing, threshold, metric, missing-**prediction** policy and test membership. Record the version before opening test labels.

A classroom result is **evidence** about a small chosen set, not approval for deployment.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines final testing and honest **claim**s.

Explain the added worked example · 2

A frozen classifier is tested on 20 fictional cases: TN 12 FP 2 FN 3 TP 3, blocked positive. Accuracy is 15/20=75 percent, precision 3/5=60 percent and recall 3/6=50 percent.

Its previously fitted always-clear baseline gets 14/20=70 percent. The candidate improves accuracy here but misses half the blocked cases.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger **claim**.

Keep the distinction between an observed result and a proposed next step.

Key terms

Term	Meaning
Freeze	Record a fixed model and procedure before testing
Held out	Kept separate from development decisions
Provenance	A record of where data and results came from
Claim	A statement whose scope must match its evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Final testing and honest claims at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Before labels	Freeze version and rules	Record test membership
During evaluation	Keep every planned case	Count failures explicitly
After evaluation	Publish full results	Restrict the claim

The test comparison must use the same cases and consistent rules. Report the denominator, reference-label process and sampling limitations. If the software fails on one case, report that failure; do not silently drop the case and keep the original headline denominator.

Why does 75 percent versus 70 percent not settle deployment? Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Hold Out Test Data

Open the final hold-out only after recording the frozen model choice.

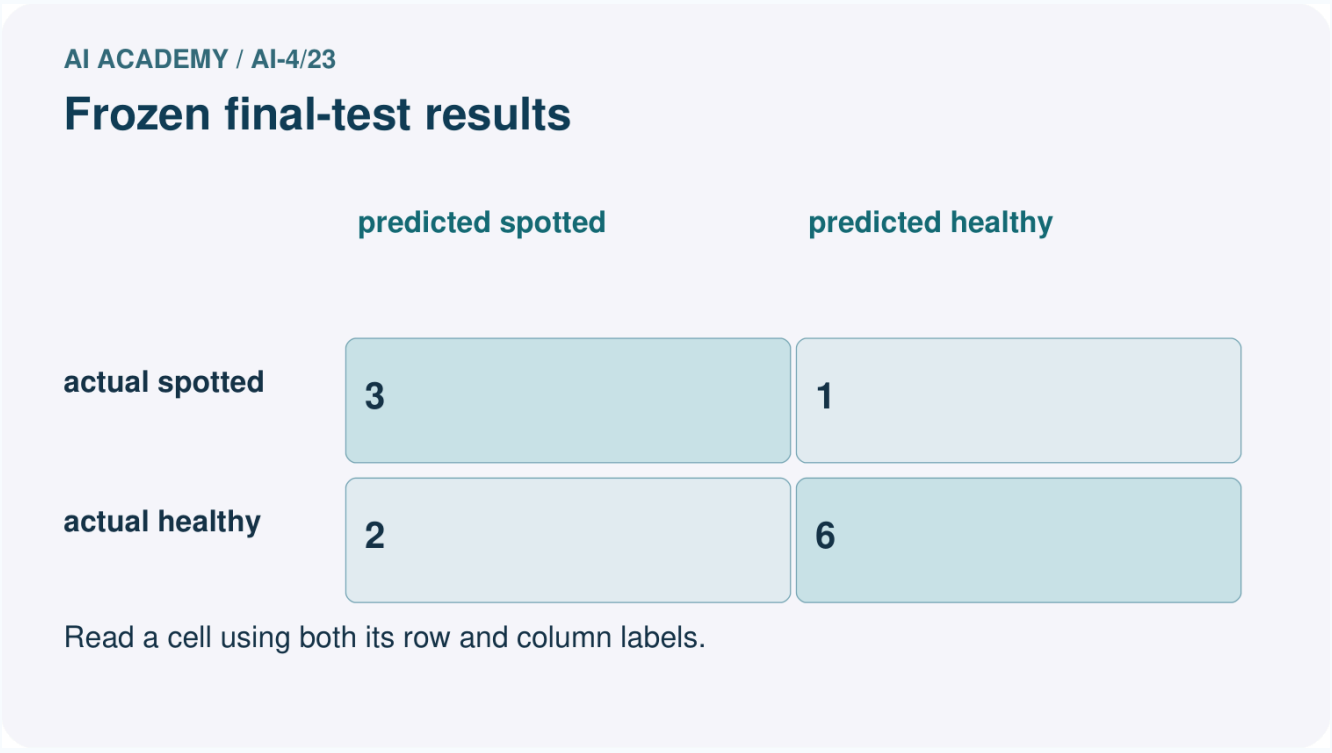
The facts we will use

Supplied facts

Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/23

Hold Out Test Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record freeze

Record model version and choices before reading these final references.



2. Calculate metrics

Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.



3. Bound claim

Report these twelve held-out cases and their collection conditions; preserve all three error records.

CHANGE ONE THING

The team changes a threshold after inspecting the false negatives.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record freeze	Record model version and choices before reading these final references.
2. Calculate metrics	Accuracy9/12=75%; precision3/5=60%; recall3/4=75%.
3. Bound claim	Report these twelve held-out cases and their collection conditions; preserve all three error records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

Change one thing in the pictured case

Change one thing

Change: The team changes a threshold after inspecting the false negatives.

Predicted result

Prediction: This set is now development **evidence** for the revised model.

Why the result follows

Reason: Using final outcomes to change the model breaks the untouched hold-out role.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Hold Out Test Data, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A final-test folder has twenty leaves; its labels are opened weekly to choose models.

2. State the decision to explain

Write the question to explain: Is it still untouched?

3. Reveal the worked reasoning

Reveal after **prediction**s: No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

Hold back the independent task

Keep for an independent attempt: You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for **claim**s.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: The difference is one correct case on 20; error consequences, limited coverage and missed blocked cases remain.

Exit response: Retain the planned denominator and report the failed case and predefined scoring policy; disclose any metric denominator change.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Why is validation not an untouched final test? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify freeze in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: calculate a final report; compare a baseline. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between freeze and held out. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: The team changes a threshold after reading test errors. How should it label later results?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Freeze the full evaluation procedure before testing; Report a held-out result with counts and limitations. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Why does 75 percent versus 70 percent not settle deployment?

Reasoning to listen for

Teacher check: Teacher explanation: The difference is one correct case on 20; error consequences, limited coverage and missed blocked cases remain.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Final testing and honest **claim**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

A newly reserved final set with appropriate grouping and fixed scoring rules, evaluated after the revision is frozen. Changed-case check: This set is now development **evidence** for the revised model.

Using final outcomes to change the model breaks the untouched hold-out role.

Changed-case reasoning: This set is now development **evidence** for the revised model. Using final outcomes to change the model breaks the untouched hold-out role.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Hold Out Test Data investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Split-Integrity Court: classify workflow actions as allowed or contaminating, then redesign the **evidence** trail.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR Test data remains unbiased as long as the model never directly fits its labels.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

A stronger **claim** makes a stronger project: Retain the planned denominator and report the failed case and predefined scoring policy; disclose any metric denominator change.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Accuracy 9/12=75 percent; precision 2/3 about 66.7 percent; recall 2/4=50 percent.

These describe these 12 held-out examples only.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Hold Out Test Data questions, teacher keys and criteria after the relevant attempts.

> Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

No. It functions as validation. Reserve a genuinely unused set for final evaluation or disclose the limitation.

Independent question

Independent question · withhold the key

You inspect final test failures, change features to fix them and report the new score on that same test set as untouched. What is wrong?

Expected independent answer

Teacher key · Expected independent answer

The test influenced development and is no longer untouched. Report this use honestly and obtain a separate final evaluation set if a new independent estimate is needed.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported **claim**s regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this **claim**: “Renaming reused data test restores its independence.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

It became validation/development **evidence**. Without fresh data, disclose that no untouched final evaluation is available and report the reused-set score as development performance.

Do not relabel it unseen because the model did not directly train on those rows.

Explanations for the additional workbook examples

W1 Calculate a final report

Workbook question · W1

Calculate the worked accuracy, precision and recall with denominators.

Teacher reasoning · W1

Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

W2 Compare a baseline

Workbook question · W2

Why does 75 percent versus 70 percent not settle deployment?

Teacher reasoning · W2

Teacher explanation: The difference is one correct case on 20; error consequences, limited coverage and missed blocked cases remain.

W3 Handle a revision

Workbook question · W3

The team changes a threshold after reading test errors. How should it label later results?

Teacher reasoning · W3

Teacher explanation: Post-test development on an inspected set, with the first frozen test retained; a fresh independent set is needed for another untouched-test **claim**.

W4 Apply independently

Workbook question · W4

A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

Teacher reasoning · W4

Teacher explanation: Accuracy 9/12=75 percent; precision 2/3 about 66.7 percent; recall 2/4=50 percent. These describe these 12 held-out examples only.

W5 Report a failure

Workbook question · W5

One planned case returns no **prediction**. What must the report retain?

Teacher reasoning · W5

Teacher explanation: Retain the planned denominator and report the failed case and predefined scoring policy; disclose any metric denominator change.

Assess independent transfer

Independent question · withhold the key

Selected case: A new frozen test has TN 7 FP 1 FN 2 TP 2. Calculate accuracy, precision and recall and state a bounded **claim**.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Accuracy 9/12=75 percent; precision 2/3 about 66.7 percent; recall 2/4=50 percent. These describe these 12 held-out examples only.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Hold Out Test Data to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Seal the Plant Classifier final test IDs and labels; record who may open them and at which release gate.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 24: Train, Validation, Test. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **freeze** and **held out** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Accuracy 9/12=75 percent; precision 2/3 about 66.7 percent; recall 2/4=50 percent. These describe these 12 held-out examples only.

Fresh retry TN 8 FP 2 FN 1 TP 1: accuracy 9/12=75 percent, precision 1/3, recall 1/2. Require counts and a scope-limited statement. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Over**fitting** and model capacity. Carry forward the saved **evidence** and the distinction between development choices and justified **claim**s.

Plan the delayed return

Suggested return points in this level: ai-4/24, ai-4/26, ai-4/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a **claim**ed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Hold Out Test Data lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: **Freeze** the full evaluation procedure before testing

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Why does 75 percent versus 70 percent not settle deployment?

Supported practice answer

Reasoned practice answer: Teacher explanation: The difference is one correct case on 20; error consequences, limited coverage and missed blocked cases remain.

Check every learner before discussion: Use the worked case for Final testing and honest **claim**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The team changes a threshold after inspecting the false negatives. Require a **prediction** and a reason before comparing. Expected result: This set is now development **evidence** for the revised model.

Reason: Using final outcomes to change the model breaks the untouched hold-out role.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

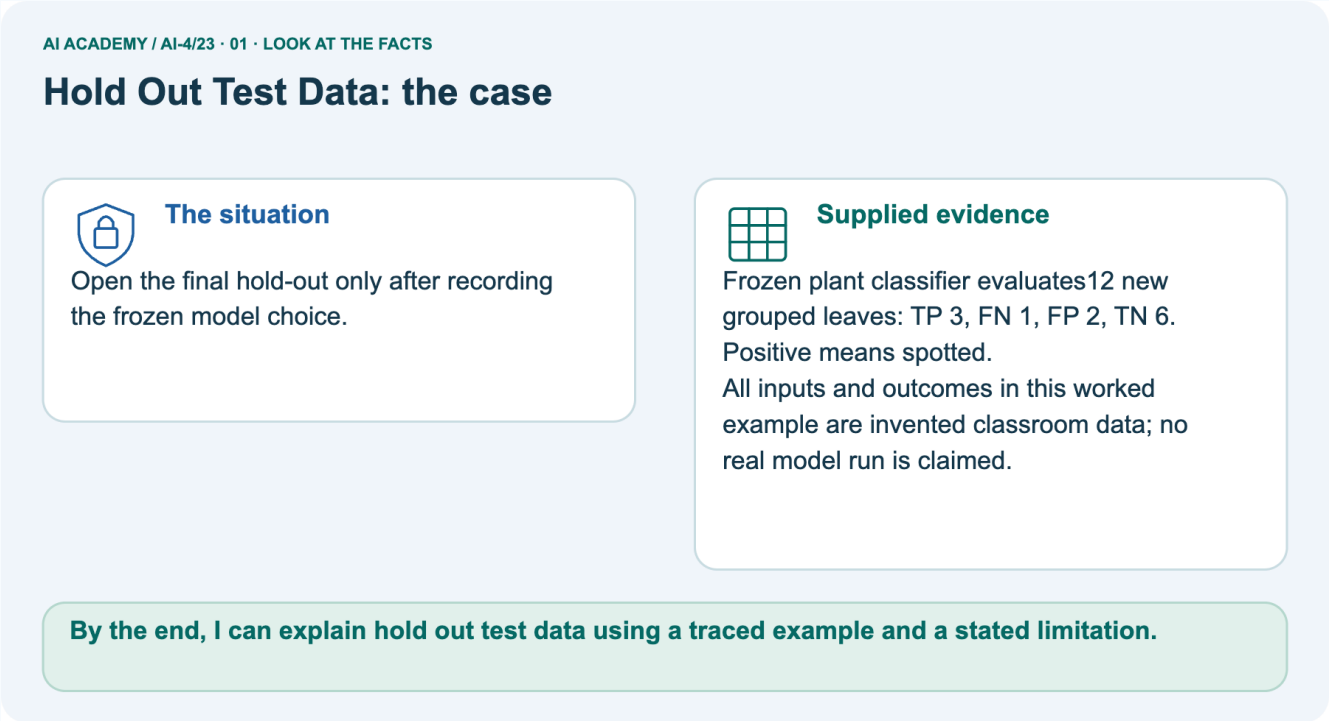
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Hold Out Test Data graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Hold Out Test Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Open the final hold-out only after recording the frozen model choice. Supplied facts: Frozen plant classifier evaluates12 new grouped leaves: TP 3, FN 1, FP 2, TN 6. Positive means spotted.

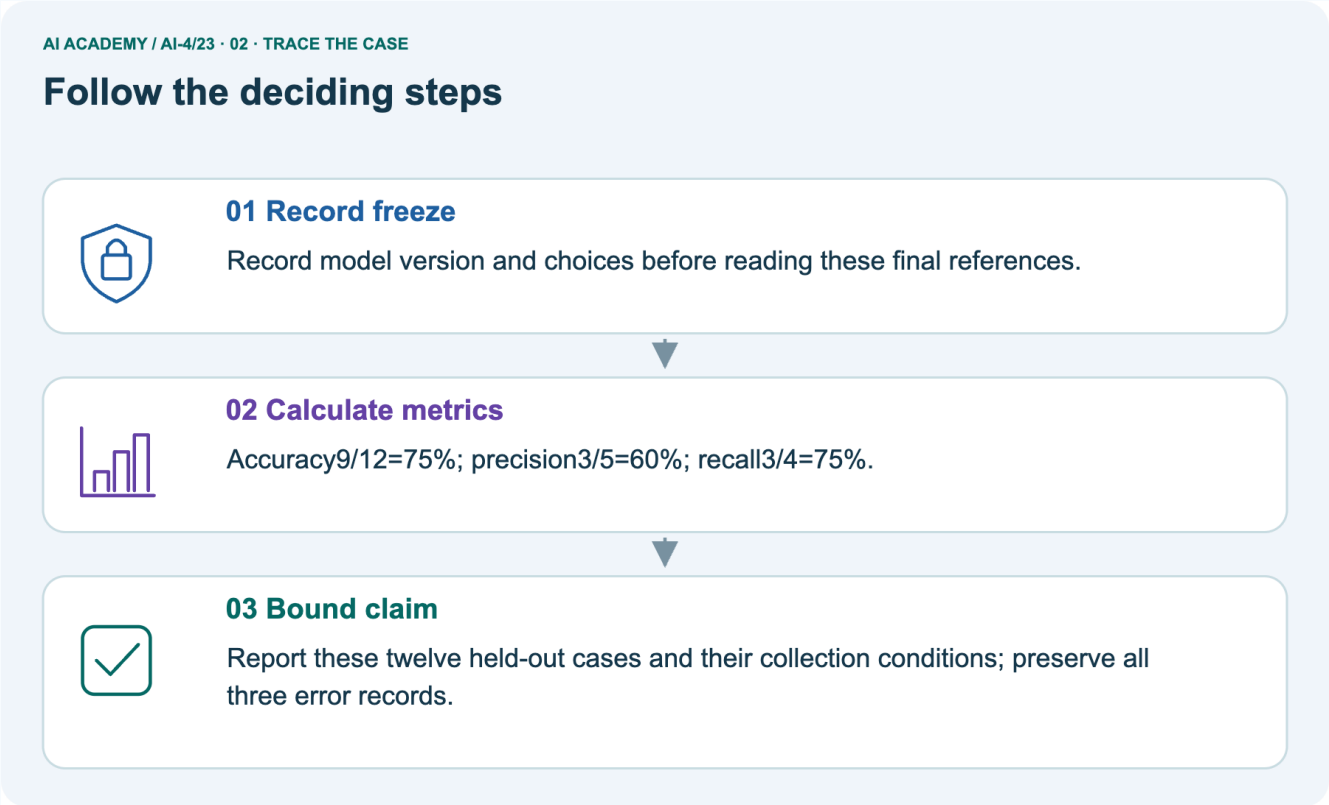
All **input**s and outcomes in this worked example are invented classroom data; no real model run is **claim**ed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

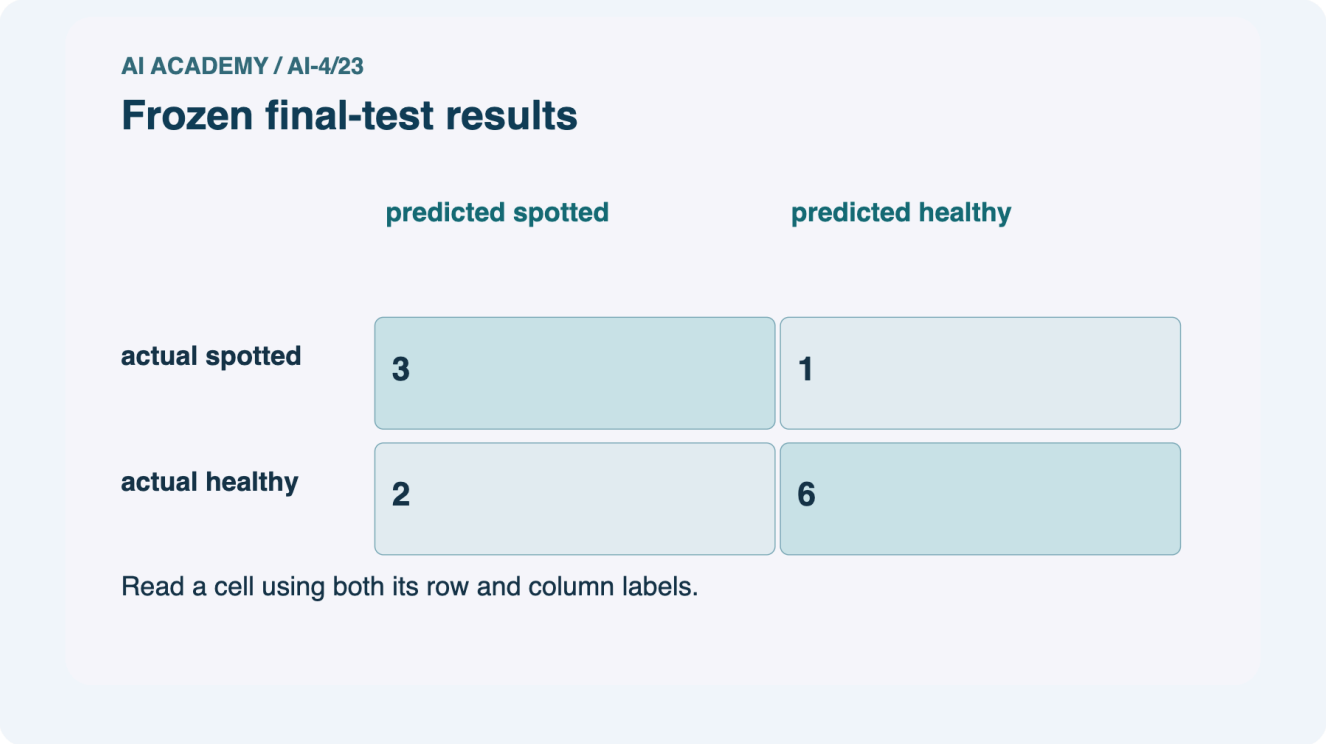
Record **freeze**: Record model version and choices before reading these final references. → Calculate metrics: Accuracy9/12=75%; precision3/5=**60%**; recall3/4=75%. → Bound **claim**: Report these twelve held-out cases and their collection conditions; preserve all three error records.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Frozen final-test results



Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen final-test results. actual spotted: predicted spotted = 3, predicted healthy = 1. actual healthy: predicted spotted = 2, predicted healthy = 6.

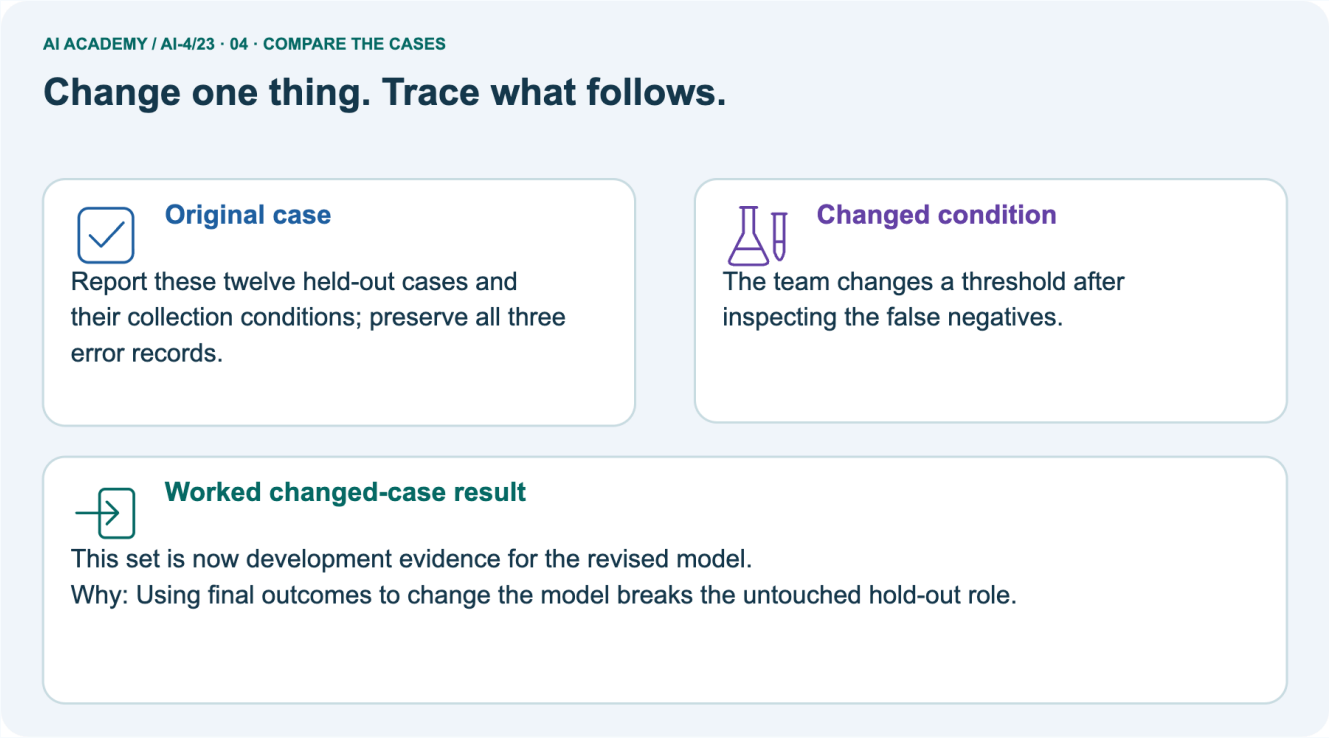
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report these twelve held-out cases and their collection conditions; preserve all three error records. Changed condition: The team changes a threshold after inspecting the false negatives.

Expected result: This set is now development **evidence** for the revised model. Reason: Using final outcomes to change the model breaks the untouched hold-out role.

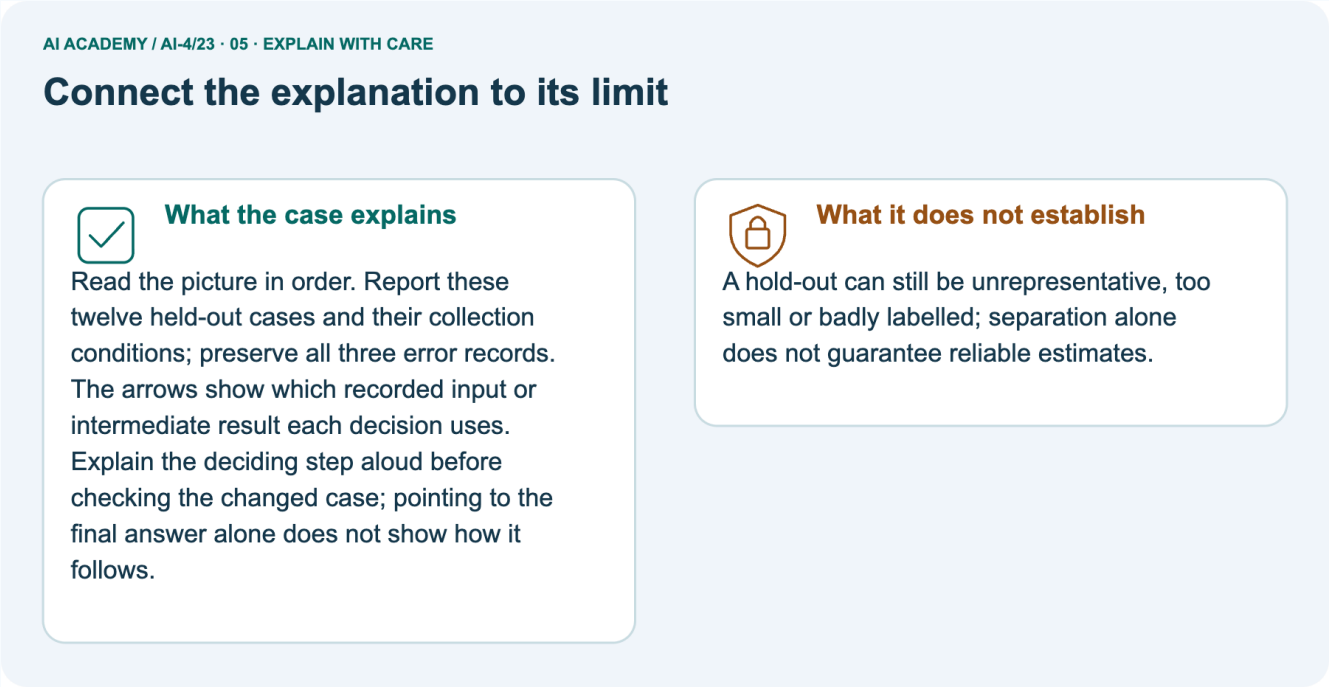
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Using final outcomes to change the model breaks the untouched hold-out role.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Report these twelve held-out cases and their collection conditions; preserve all three error records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

Explain the case, then point to a limit on the **claim**.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the **claim**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A hold-out can still be unrepresentative, too small or badly labelled; separation alone does not guarantee reliable estimates.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is needed for an honest final **claim** about the revised version? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A newly reserved final set with appropriate grouping and fixed scoring rules, evaluated after the revision is frozen. Changed-case check: This set is now development **evidence** for the revised model.

Using final outcomes to change the model breaks the untouched hold-out role.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.